Quality-aware and Soft Consistency Driven Representation Fusion for Incomplete Multi-view Multi-label Classification

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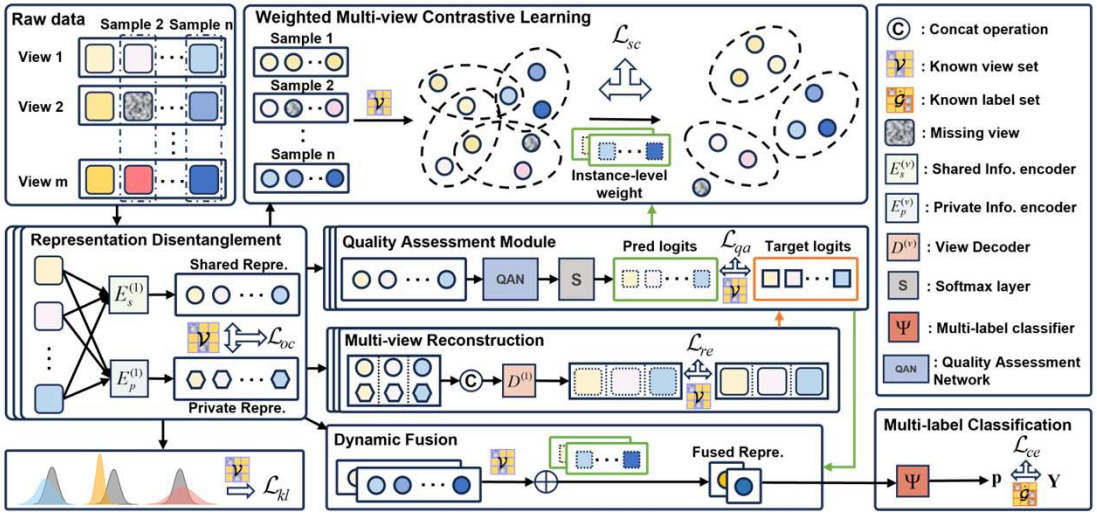
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Method

- **Motivation:** the contrastive learning paradigm faces a dual challenge under incomplete data settings. Traditional methods use strong consistency constraints, which force the model to discard view-specific information that is discriminative for classification. Furthermore, existing fusion strategies are often static and treat all samples equally, failing to adapt to the varying quality of different sample views.

- **Components:** We propose a quality-aware representation learning via soft consistency framework, named QUALITY. It integrates four core modules: a representation disentanglement module that decomposes representations; a soft consistency constraint module that aligns shared information while preserving task-relevant details; a quality assessment module that predicts instance-level quality scores for dynamic fusion; and a multi-label classification module that predicts the final labels.



Contribution

- We propose a soft consistency constraint that aligns shared information across views but prevents the over-compression of view-specific private information.
- We design a Quality Assessment Network (QAN) that achieves instance-level dynamic fusion, assigning adaptive weights based on data quality rather than using static strategies.
- Our method shows good adaptability to arbitrary view and label incompleteness. Extensive experimental results, both in complete and incomplete cases, demonstrate the superiority and robustness of our method.

Experimental Results

Data	Metric	CDMM	DM2L	LVSL	iMVWL	NAIM3L	DICNet	DIMC	MSLPP	SIP	QUALITY
Corel5K	AP	$0.354_{0.004}$	$0.262_{0.005}$	$0.342_{0.004}$	$0.283_{0.008}$	$0.309_{0.004}$	$0.381_{0.004}$	$0.353_{0.006}$	$0.413_{0.008}$	$0.418_{0.009}$	$0.434_{0.010}$
	1-HL	$0.987_{0.000}$	$0.987_{0.000}$	$0.987_{0.000}$	$0.978_{0.000}$	$0.987_{0.000}$	$0.988_{0.000}$	$0.987_{0.000}$	$0.988_{0.000}$	$0.988_{0.000}$	$0.988_{0.000}$
	1-RL	$0.884_{0.003}$	$0.843_{0.002}$	$0.881_{0.003}$	$0.865_{0.005}$	$0.878_{0.002}$	$0.882_{0.004}$	$0.867_{0.001}$	$0.901_{0.003}$	$0.911_{0.003}$	$0.913_{0.003}$
	AUC	$0.888_{0.003}$	$0.845_{0.002}$	$0.884_{0.003}$	$0.868_{0.005}$	$0.881_{0.002}$	$0.884_{0.004}$	$0.870_{0.001}$	$0.903_{0.004}$	$0.913_{0.003}$	$0.915_{0.003}$
	1-OE	$0.410_{0.007}$	$0.295_{0.014}$	$0.391_{0.009}$	$0.311_{0.015}$	$0.350_{0.009}$	$0.468_{0.007}$	$0.422_{0.015}$	$0.485_{0.001}$	$0.489_{0.016}$	$0.511_{0.018}$
	1-Cov	$0.723_{0.007}$	$0.647_{0.005}$	$0.718_{0.006}$	$0.702_{0.008}$	$0.725_{0.005}$	$0.727_{0.011}$	$0.684_{0.011}$	$0.766_{0.009}$	$0.787_{0.009}$	$0.795_{0.008}$
Pascal07	AP	$0.508_{0.005}$	$0.471_{0.008}$	$0.504_{0.005}$	$0.437_{0.018}$	$0.488_{0.003}$	$0.505_{0.012}$	$0.532_{0.002}$	$0.544_{0.010}$	$0.555_{0.010}$	$0.590_{0.009}$
	1-HL	$0.931_{0.001}$	$0.928_{0.001}$	$0.930_{0.000}$	$0.882_{0.004}$	$0.928_{0.001}$	$0.929_{0.001}$	$0.931_{0.001}$	$0.932_{0.001}$	$0.931_{0.001}$	$0.934_{0.001}$
	1-RL	$0.812_{0.004}$	$0.761_{0.005}$	$0.806_{0.003}$	$0.736_{0.015}$	$0.783_{0.001}$	$0.783_{0.008}$	$0.813_{0.000}$	$0.819_{0.006}$	$0.830_{0.004}$	$0.852_{0.005}$
	AUC	$0.838_{0.003}$	$0.779_{0.004}$	$0.832_{0.002}$	$0.767_{0.015}$	$0.811_{0.001}$	$0.809_{0.006}$	$0.833_{0.002}$	$0.841_{0.004}$	$0.850_{0.005}$	$0.870_{0.004}$
	1-OE	$0.419_{0.008}$	$0.420_{0.011}$	$0.419_{0.008}$	$0.362_{0.023}$	$0.421_{0.006}$	$0.427_{0.015}$	$0.456_{0.011}$	$0.466_{0.014}$	$0.464_{0.018}$	$0.499_{0.013}$
	1-Cov	$0.759_{0.003}$	$0.692_{0.004}$	$0.751_{0.003}$	$0.677_{0.015}$	$0.727_{0.002}$	$0.731_{0.006}$	$0.769_{0.007}$	$0.771_{0.003}$	$0.783_{0.006}$	$0.808_{0.006}$
ESPGame	AP	$0.289_{0.003}$	$0.212_{0.002}$	$0.285_{0.003}$	$0.244_{0.005}$	$0.246_{0.002}$	$0.297_{0.002}$	$0.287_{0.002}$	$0.310_{0.004}$	$0.311_{0.004}$	$0.318_{0.004}$
	1-HL	$0.983_{0.000}$	$0.982_{0.000}$	$0.983_{0.000}$	$0.972_{0.000}$	$0.983_{0.000}$	$0.983_{0.000}$	$0.983_{0.000}$	$0.983_{0.000}$	$0.983_{0.000}$	$0.983_{0.000}$
	1-RL	$0.832_{0.001}$	$0.781_{0.001}$	$0.829_{0.001}$	$0.808_{0.002}$	$0.818_{0.002}$	$0.832_{0.001}$	$0.821_{0.000}$	$0.843_{0.002}$	$0.849_{0.002}$	$0.854_{0.002}$
	AUC	$0.836_{0.001}$	$0.785_{0.001}$	$0.833_{0.002}$	$0.813_{0.002}$	$0.824_{0.002}$	$0.836_{0.001}$	$0.826_{0.000}$	$0.847_{0.002}$	$0.853_{0.002}$	$0.858_{0.002}$
	1-OE	$0.396_{0.005}$	$0.294_{0.006}$	$0.389_{0.004}$	$0.343_{0.013}$	$0.339_{0.003}$	$0.439_{0.007}$	$0.435_{0.009}$	$0.457_{0.012}$	$0.455_{0.007}$	$0.461_{0.008}$
	1-Cov	$0.574_{0.004}$	$0.488_{0.003}$	$0.567_{0.005}$	$0.548_{0.004}$	$0.571_{0.003}$	$0.593_{0.003}$	$0.562_{0.004}$	$0.622_{0.005}$	$0.628_{0.005}$	$0.641_{0.004}$
IAPRTC12	AP	$0.305_{0.004}$	$0.234_{0.003}$	$0.304_{0.004}$	$0.237_{0.003}$	$0.261_{0.001}$	$0.323_{0.001}$	$0.308_{0.001}$	$0.340_{0.005}$	$0.331_{0.006}$	$0.347_{0.004}$
	1-HL	$0.981_{0.000}$	$0.980_{0.000}$	$0.981_{0.000}$	$0.969_{0.000}$	$0.980_{0.000}$	$0.981_{0.000}$	$0.980_{0.000}$	$0.981_{0.000}$	$0.980_{0.000}$	$0.981_{0.000}$
	1-RL	$0.862_{0.002}$	$0.823_{0.002}$	$0.861_{0.002}$	$0.833_{0.002}$	$0.848_{0.001}$	$0.873_{0.001}$	$0.864_{0.000}$	$0.882_{0.002}$	$0.885_{0.003}$	$0.893_{0.002}$
	AUC	$0.864_{0.002}$	$0.825_{0.001}$	$0.863_{0.001}$	$0.835_{0.001}$	$0.850_{0.001}$	$0.874_{0.000}$	$0.864_{0.000}$	$0.883_{0.002}$	$0.886_{0.002}$	$0.894_{0.002}$
	1-OE	$0.432_{0.008}$	$0.340_{0.006}$	$0.429_{0.009}$	$0.352_{0.008}$	$0.390_{0.005}$	$0.468_{0.002}$	$0.431_{0.006}$	$0.474_{0.008}$	$0.463_{0.009}$	$0.481_{0.007}$
	1-Cov	$0.597_{0.004}$	$0.529_{0.004}$	$0.597_{0.004}$	$0.564_{0.005}$	$0.592_{0.004}$	$0.649_{0.001}$	$0.597_{0.004}$	$0.672_{0.006}$	$0.675_{0.007}$	$0.693_{0.006}$
MirFlickr	AP	$0.570_{0.002}$	$0.514_{0.006}$	$0.553_{0.002}$	$0.490_{0.012}$	$0.551_{0.002}$	$0.589_{0.005}$	$0.602_{0.002}$	$0.615_{0.004}$	$0.614_{0.004}$	$0.633_{0.004}$
	1-HL	$0.886_{0.001}$	$0.878_{0.001}$	$0.885_{0.001}$	$0.839_{0.002}$	$0.882_{0.001}$	$0.888_{0.002}$	$0.888_{0.000}$	$0.892_{0.001}$	$0.891_{0.001}$	$0.895_{0.001}$
	1-RL	$0.856_{0.001}$	$0.831_{0.003}$	$0.856_{0.001}$	$0.803_{0.008}$	$0.844_{0.001}$	$0.863_{0.004}$	$0.865_{0.001}$	$0.879_{0.002}$	$0.877_{0.002}$	$0.886_{0.002}$
	AUC	$0.846_{0.001}$	$0.828_{0.003}$	$0.844_{0.001}$	$0.787_{0.012}$	$0.837_{0.001}$	$0.849_{0.004}$	$0.852_{0.001}$	$0.865_{0.002}$	$0.860_{0.003}$	$0.872_{0.002}$
	1-OE	$0.631_{0.004}$	$0.510_{0.008}$	$0.607_{0.004}$	$0.511_{0.022}$	$0.585_{0.003}$	$0.637_{0.007}$	$0.647_{0.007}$	$0.667_{0.007}$	$0.662_{0.008}$	$0.682_{0.006}$
	1-Cov	$0.640_{0.001}$	$0.604_{0.005}$	$0.636_{0.001}$	$0.572_{0.013}$	$0.631_{0.002}$	$0.652_{0.007}$	$0.661_{0.003}$	$0.679_{0.003}$	$0.678_{0.003}$	$0.695_{0.003}$

Table 1: Experimental results of ten methods on five datasets with 50% missing-view rate and 50% missing-label rate (the bottom right digit is the standard deviation).

Strategies	Dataset	AP	1-HL	1-RL	AUC
Pred Fusion	Corel5K	0.434	0.988	0.913	0.915
Direct Fusion	Corel5K	0.426	0.987	0.910	0.912
Pred Fusion	Pascal07	0.590	0.934	0.852	0.870
Direct Fusion	Pascal07	0.581	0.933	0.848	0.866

Table 2: Performance of our method under different fusion strategies, where 'Pred Fusion' uses the weights output by the quality assessment network for fusion, while 'Direct Fusion' uses the normalized negative exp-loss for fusion.

Backbone	$\mathcal{L}_{oc}$	$\mathcal{L}_{qa}$	Corel5k AP	Corel5k AUC	Pascal07 AP	Pascal07 AUC
✓			0.3406	0.8801	0.5544	0.8522
✓	✓		0.3691	0.8906	0.5736	0.8621
✓		✓	0.4057	0.9081	0.5708	0.8605
✓	✓	✓	0.4335	0.9153	0.5901	0.8696

Table 3: The ablation experimental results on the Corel5K and Pascal07 datasets, and all datasets are with 50% missing views, 50% missing labels and 70% training samples.

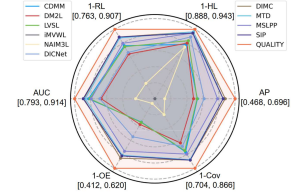


Figure 3: The experimental results of nine methods on the complete Pascal07 dataset. In this visualization, the worst results are positioned at the center, while the best results correspond to the vertices, based on six evaluation metrics.

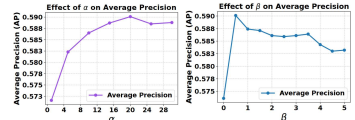


Figure 4: Hyper-parameters analysis regarding α and β on the Pascal07 dataset (with a setting of 50% view and 50% label double missingness).

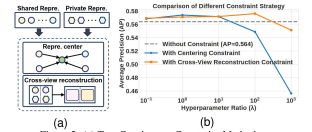


Figure 2: (a) Two Consistency Constraint Methods. (b) Performance of the Two Constraints with Varying Hyper-parameters on Pascal07 dataset.